Pandas Profiling: A Full EDA Report in One Line

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Where this fits: Pipeline map step 3 (Understand data). See the Course map.

0 Foundations	1 Frame	2 Get data	**3 Understand**	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Exploratory data analysis; Setup: conda, Jupyter and Colab.
- **Compare with:** Bivariate and multivariate analysis; Univariate analysis.

1. Overview

Key point: A profiling library runs most of the hand-made EDA (first questions, univariate, bivariate and multivariate analysis) by itself and writes the results into one web page.

So far we built EDA by hand, in three steps: the first questions for a new dataset, **univariate analysis** (G-2050), and bivariate and multivariate analysis. Each step took a few lines of pandas or one plot at a time. A **profiling report** does most of this work in one go: we hand it a DataFrame, and it returns a web page that describes every column and every pair of columns.

The library that builds it is known as **Pandas Profiling** (G-1440). Figure 1 shows what one call of `ProfileReport` produces: a report with six sections, each answering questions of the hand-made EDA.

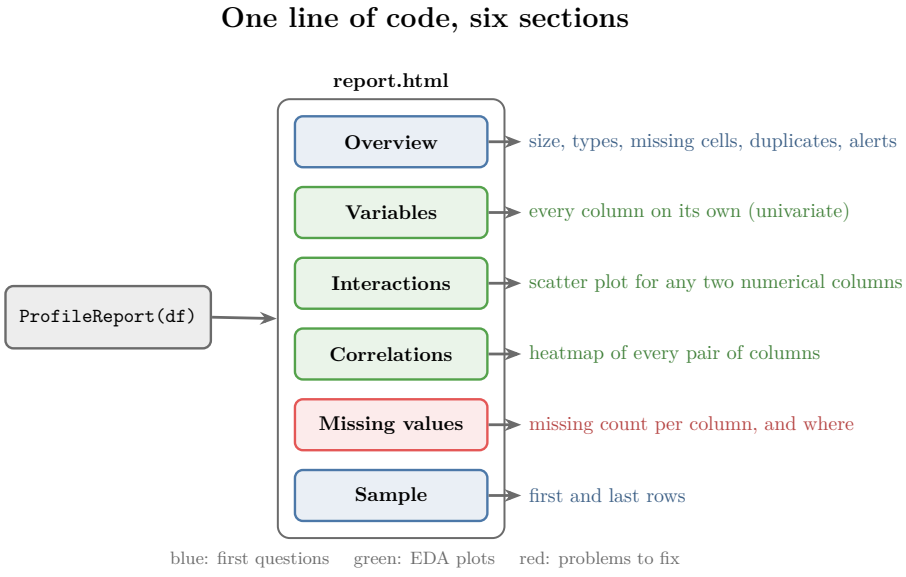


Figure 1: What a profiling report contains

We use the same Titanic data as before, 891 passengers. The Notebook (`ML-021-pandas-profiling.ipynb`) builds the report and saves it as `titanic_report.html`; Figures 3 to 11 are screenshots of that page.

2. Building the report

> **Key point:** Install the library once; then three lines build the report and save it as an **HTML** (G-904) file.

2.1 Installing the library

> **Key point:** One `pip install` command adds the library to our Python setup.

Pandas Profiling (G-1440) is not part of pandas; it is a separate library. We install it once, from a terminal or from a notebook cell. The command downloads the library from **PyPI** (G-1593), the public store of Python libraries, and installs it on our machine.

> **Python:** Installing the library.
> `pip install fg-data-profiling`
> In a Jupyter cell, put `!` in front: `!pip install fg-data-profiling`. The `!` sends the line to the terminal instead of to Python.

> **Extra:** Version clash (October 2026). The library still requires pandas older than 3.0. In a setup that already has pandas 3, a plain `pip install fg-data-profiling` quietly downgrades pandas (and NumPy, SciPy) to older versions. To keep pandas 3, install it with `pip install --no-deps fg-data-profiling` and then install the few small helper libraries it asks for when imported. The report then works with the one-line fix in Section 2.2. In a fresh setup made just for profiling, the plain command is fine.

> **Extra:** The library has changed its name twice. The library began as `pandas-profiling` (import `pandas_profiling`). In 2023 it became `ydata-profiling` (import `ydata_profiling`), and in 2026 `fg-data-profiling` (import `data_profiling`). The old names still appear in many tutorials; `pip install pandas-profiling` now installs only an empty package that says to use the new name, and `import ydata_profiling` warns that it gets no more updates (PyPI pages of the three packages). The report itself and the `ProfileReport` call are the same under all three names.

2.2 Three lines of code

> **Key point:** `ProfileReport(df)` builds the report; `to_file` saves it as a web page we open in any browser.

Once installed, building a report takes three steps (Figure 2):

1. Import `ProfileReport` from the library.
2. Call `ProfileReport` on our DataFrame. The call creates a report object.
3. Save the object as an **HTML** (G-904) file with `to_file`.

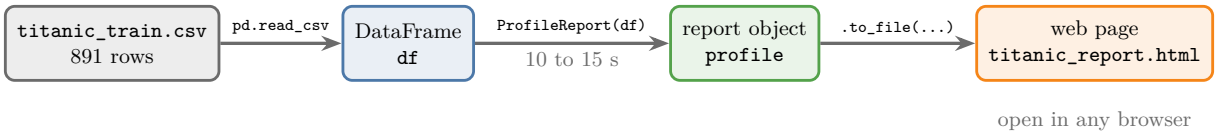


Figure 2: The three lines of code as a flow: the CSV file becomes a DataFrame, `ProfileReport` builds the report object in 10 to 15 seconds, and `to_file` saves it as a web page

> **Python:** Building and saving the report.
> ```
> import pandas as pd
> from data_profiling import ProfileReport
>
> df = pd.read_csv("data/titanic_train.csv")
> profile = ProfileReport(df, title="Titanic passengers")
> profile.to_file("titanic_report.html")
> ```
> `title` is optional; it becomes the heading of the page. Double-clicking the HTML file opens the report in a browser.

The report takes a few seconds to build. The time grows with the size of the data: behind the scenes, the library computes statistics and draws charts for every column and every pair of columns. For the 891 Titanic rows it takes about ten to fifteen seconds.

> **Extra:** Two more options are useful:
> - `ProfileReport(df, minimal=True)` skips the slow parts (correlations, interactions, missing-value charts). On data with many thousands of rows or many columns, the full report can take minutes; the minimal one stays fast.
> - `profile.to_notebook_iframe()` shows the report inside Jupyter instead of saving a file, in an **iframe** (a frame that shows one web page inside another). The iframe stores the whole page in the notebook, which makes the notebook file several megabytes big.

> **Extra:** The library officially supports pandas versions below 3. With pandas 3, building the report fails with the error `'ArrowExtensionArray' object has no attribute 'sum'`. pandas 3 stores text columns in a new format (Arrow arrays) that the library cannot add up. One line before building the report switches text back to plain Python strings:
> ```
> pd.set_option("mode.string_storage", "python")
> ```

The page has six sections, listed in Figure 1. A menu at the top jumps to each one. The sections follow the order of the hand-made EDA: first the basic questions, then one column at a time, then pairs of columns.

3. Overview: the first questions, answered

> **Key point:** The Overview section answers the first questions about a new dataset (size, types, missing values, duplicates) in one table.

3.1 Dataset statistics

> **Key point:** Rows, columns, missing cells, duplicate rows and memory use, all in one table.

The Overview section opens with the dataset as a whole. Figure 3 shows it for the Titanic data. These are the same first questions we asked with `df.shape`, `df.info()`, `df.isnull().sum()` and `df.duplicated().sum()` in how big is the data.

Dataset statistics

Number of variables	12
Number of observations	891
Missing cells	866
Missing cells (%)	8.1%
Duplicate rows	0
Duplicate rows (%)	0.0%
Total size in memory	118.9 KiB
Average record size in memory	136.6 B

Variable types

Numeric	5
Categorical	4
Text	3

Figure 3: The Overview section: dataset statistics and variable types

Reading the left table from the top:

- **Number of variables:** 12. The report calls columns **variables**.
- **Number of observations:** 891. The report calls rows **observations**: one record each, here one passenger.
- **Missing cells:** 866, which is 8.1% of all cells.
- **Duplicate rows:** 0, so no passenger appears twice.
- **Total size in memory:** 118.9 KiB for the whole table; the **average record size** is 136.6 bytes, the memory one row takes.

The missing-cells percentage is worked out from the whole grid of cells.

> **Extra:** The share of missing cells, step by step.
>
> 1. **In words:** count all cells (rows times columns), then divide the missing cells by that total.

2. **Formula:**

$$\text{all cells} = \text{rows} \times \text{columns}$$

$$\text{missing percent} = \frac{\text{missing cells}}{\text{all cells}} \times 100$$

3. **With the Titanic numbers:** the grid has this many cells:

$$891 \times 12 = 10{,}692$$

The share of missing cells is then:

$$\frac{866}{10{,}692} \times 100 = 8.1 \text{ percent}$$

The 866 missing cells come from three columns: 177 in `Age`, 687 in `Cabin` and 2 in `Embarked`.

The right table counts the column types. The report sorts the 12 columns into three kinds:

Type	Count	Columns
Numeric	5	`PassengerId`, `Age`, `SibSp`, `Parch`, `Fare`
Categorical	4	`Survived`, `Pclass`, `Sex`, `Embarked`
Text	3	`Name`, `Ticket`, `Cabin`

The report decides the type itself, from the values. `Survived` and `Pclass` hold numbers, but only two or three different ones, so the report treats them as categories. The report's split matches the split into numerical and categorical columns from numerical and categorical data.

Extra: "Text" is the report's name for a column of free text, where nearly every value is different (every passenger has their own name and ticket number). Older versions of the library put such columns under "Categorical" and flagged them with a **high cardinality** warning: a categorical column with very many different categories.

3.2 Alerts

Key point: Alerts are a ready-made list of possible problems, one line per column; each one is a question to check, not a verdict.

The second tab of the Overview section lists **alerts**: warnings about columns that may need attention. Figure 4 shows the twelve alerts for the Titanic data.

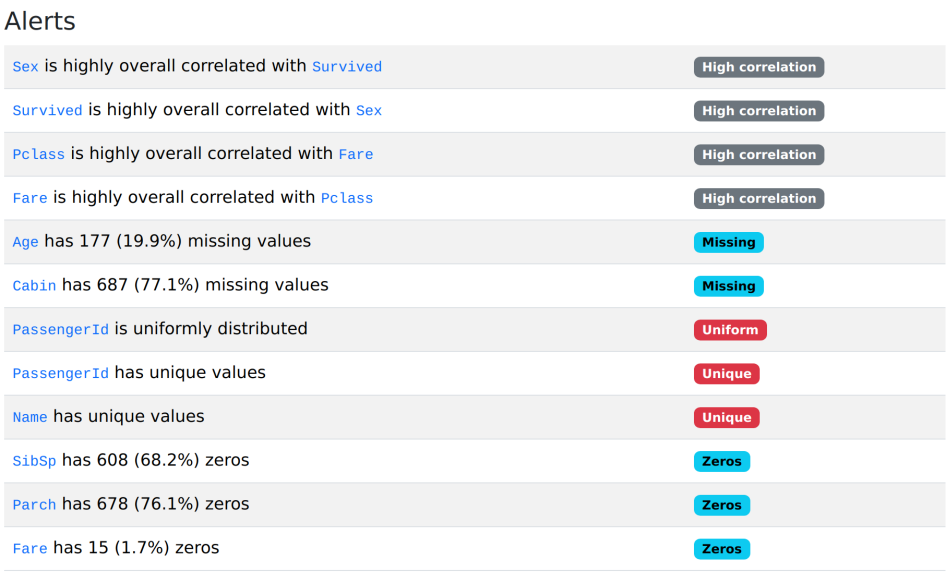


Figure 4: The Alerts tab of the Overview section

The alerts fall into five kinds:

- **High correlation:** `Sex` is strongly linked to `Survived`, and `Pclass` to `Fare`. These are the strongest relationships in the data, worth studying first.
- **Missing:** `Age` is missing for 177 passengers (19.9%) and `Cabin` for 687 (77.1%). Nearly a fifth of the ages must be dealt with before training a model. With more than three quarters missing, `Cabin` may be unusable.
- **Uniform:** every value of `PassengerId` occurs equally often (once each). A running number like this carries no information about the passenger.
- **Unique:** every value of `PassengerId` and of `Name` is different. A column that is different for every row cannot reveal a pattern shared by several rows.
- **Zeros:** most passengers have 0 in `SibSp` (68.2%) and `Parch` (76.1%): they travelled without siblings, spouses, parents or children. 15 fares (1.7%) are 0, which is odd for a paid ticket.

In seconds, the alerts point to the columns that need cleaning and the columns that look useless. Each alert still needs a judgement: many zeros in `SibSp` is simply a fact about the passengers, while 15 free tickets deserve a closer look.

> **Extra:** The third tab, Reproduction, records when the report was made, how long it took, and the library version and settings. The Reproduction tab lets someone else rebuild exactly the same report later.

4. Variables: univariate analysis of every column

> **Key point:** The Variables section gives each column its own block of statistics and a chart, chosen by the column's type.

The second section is the **univariate analysis** (G-2050) (one column at a time), done for every column at once. A drop-down menu picks one column, or shows all of them one below the other. What each block contains depends on the column's type.

Some columns are not worth studying. `PassengerId` is just a running number, as its alerts already said, so we skip it.

4.1 A categorical column: Sex

> **Key point:** For a categorical column, the report counts each category and draws the counts as bars, like a count plot.

Figure 5 shows the block for `Sex`. On the left, a table gives the basic facts; on the right, a bar for each category with its count.

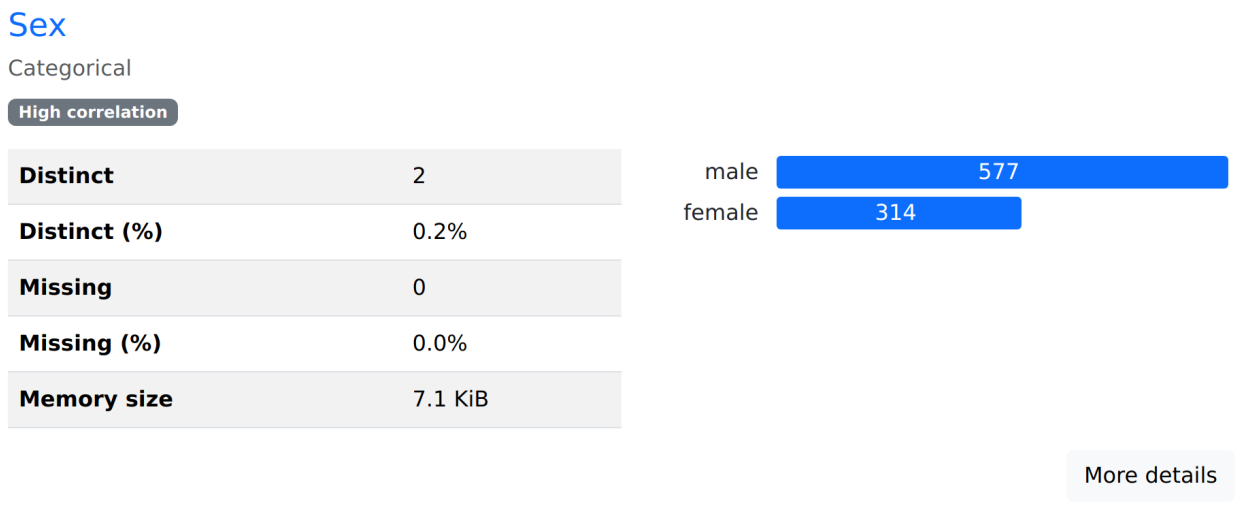


Figure 5: The Variables block for Sex

- **Distinct:** 2 different values, male and female. As a share of the 891 rows, that is 0.2%.
- **Missing:** 0, so every passenger has a recorded sex.
- **The bars:** 577 male and 314 female passengers. The bar chart is the **count plot** (G-494) (one bar per category, as tall as its count), drawn sideways.

The "More details" button opens four further tabs:

- **Overview:** the length of the values in characters (here 4 for "male", 6 for "female") and the first five values as a sample.
- **Categories:** every category with its count and percentage (64.8% male, 35.2% female).

- **Words** and **Characters:** the words and characters that make up the values, which matter more for text columns.

4.2 A numerical column: Age

> **Key point:** For a numerical column, the report gives the main summary numbers and a histogram; "More details" adds percentiles, spread and the extreme values.

Figure 6 shows the block for `Age`. The missing values are printed in red, because they triggered an alert.

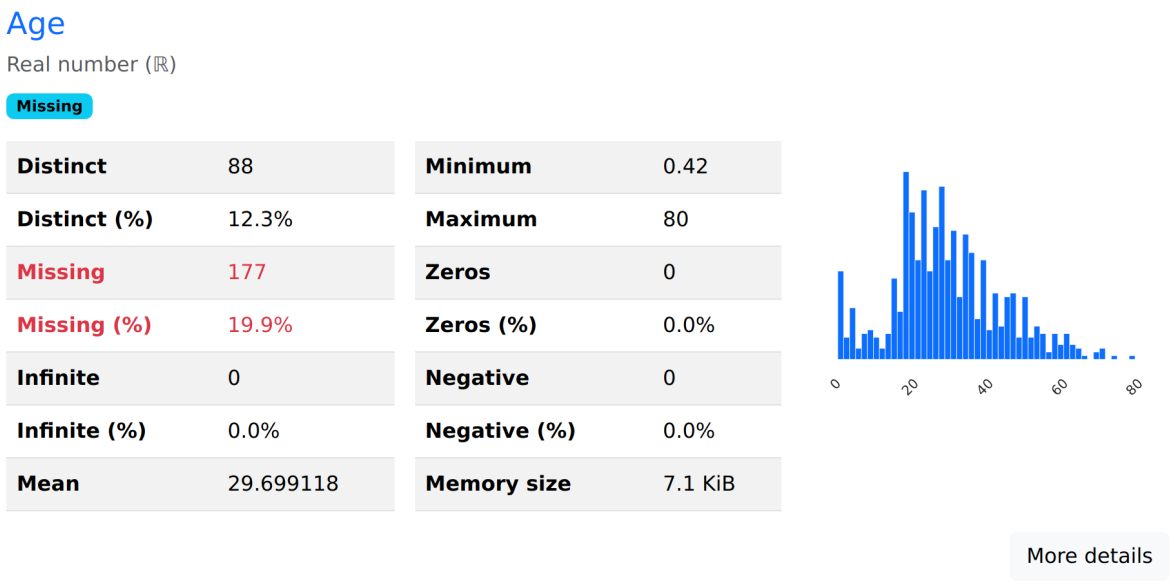


Figure 6: The Variables block for Age

The two tables give, among others:

- **Distinct:** 88 different ages.
- **Missing:** 177 (19.9%).
- **Infinite:** 0. An infinite value can appear after a division by zero, and would break most calculations.
- **Mean:** 29.70 years. **Minimum:** 0.42 (a baby of five months). **Maximum:** 80.
- **Zeros** and **Negative:** 0 each. A negative age would be an error in the data.

On the right is the histogram of `Age`, the same shape as in the histogram.

"More details" opens four tabs: Statistics, Histogram, Common values and Extreme values. The Statistics tab (Figure 7) has two tables.

Quantile statistics

Minimum	0.42
5-th percentile	4
Q1	20.125
median	28
Q3	38
95-th percentile	56
Maximum	80
Range	79.58
Interquartile range (IQR)	17.875

Descriptive statistics

Standard deviation	14.526497
Coefficient of variation (CV)	0.48912219
Kurtosis	0.17827415
Mean	29.699118
Median Absolute Deviation (MAD)	9
Skewness	0.38910778
Sum	21205.17
Variance	211.01912
Monotonicity	Not monotonic

Figure 7: The Statistics tab of Age, under More details

Quantile statistics describe positions in the sorted data:

- **5th and 95th percentile:** 4 and 56. Only 5% of the passengers were younger than 4, and only 5% older than 56.
- **Q1, median, Q3:** 20.125, 28 and 38. The **first quartile** Q1 is the age a quarter of the passengers are below; the median splits them in half; the **third quartile** Q3 is the age three quarters are below. These cut points are **quantiles**: values that a given share of the data lies below (the 5th percentile above is one too). Together with the minimum and maximum, Q1, median and Q3 give the five-number summary drawn by a box plot (G-329).
- **Range:** maximum minus minimum:

$$80 - 0.42 = 79.58$$

- **Interquartile range (IQR):** the width of the middle half of the ages, Q3 minus Q1:

$$38 - 20.125 = 17.875$$

Descriptive statistics describe the centre and the spread:

- **Standard deviation:** 14.53, and **variance:** 211.02, its square.
- **Skewness:** 0.39, slightly skewed to the right (see skewness (G-1817)).
- **Sum:** 21,205.17, all known ages added up.
- **Coefficient of variation**, **kurtosis**, **median absolute deviation** and **monotonicity** are new; the Extra boxes below explain them.

> **Extra:** The **coefficient of variation (CV)** compares the spread with the average.
> 1. **In words:** divide the standard deviation by the mean. The result has no units, so

it can compare the spread of columns measured in different units (years, rupees).

2. **Formula:**
$$\text{CV} = \frac{\text{standard deviation}}{\text{mean}}$$
3. **With the Age numbers:**
$$\text{CV} = \frac{14.53}{29.70} = 0.49$$

So ages typically lie about half the mean (around 15 years) away from the mean.

Extra: The **median absolute deviation (MAD)** is a measure of spread built from medians instead of means.

1. **In words:** find the median; measure how far each value is from it; the MAD is the median of those distances.
2. **Formula:**
$$\text{MAD} = \text{median}\Big(|x_i - \text{median}(x)| \Big)$$
3. **With five values** 1, 2, 3, 4, 10:
   - the median of the values is 3;
   - the distances from 3 are 2, 1, 0, 1, 7;
   - sorted, the distances are 0, 1, 1, 2, 7;
   - their median, the middle one, is the MAD:

$$\text{MAD} = 1$$

The far value 10 hardly changes the MAD, while it would raise the standard deviation a lot. For `Age`, the MAD is 9 years.

Extra: Two more entries in Figure 7:

- **Kurtosis** measures how heavy the tails of a distribution are compared with a normal (bell-shaped) curve. The report gives 0 for a normal curve and a positive value when extreme values are more common than in one. `Age` has 0.18, close to normal. In its simple form, with m_2 the average of the squared distances from the mean (the variance) and m_4 the average of their fourth powers:
$$\text{kurtosis} = \frac{m_4}{m_2^2} - 3$$
Two lists of ten values show the effect. Nine values of 5 and one value of 15 have a single extreme value. Step by step:
$$\text{mean} = (9 \times 5 + 15)/10 = 6$$
The distances from 6 are -1 (nine times) and 9 (once).
$$m_2 = (9 \times 1 + 81)/10 = 9$$

$$m_4 = (9 \times 1 + 6561)/10 = 657$$

$$\text{kurtosis} = 657/81 - 3 = 5.11$$

The values 1 to 10 are spread evenly and have no extreme value; the same steps (mean 5.5, $m_2 = 8.25$, $m_4 = 120.86$) give

$$120.86/8.25^2 - 3 = -1.22$$

The extreme value is what makes the number large and positive.

- **Monotonicity** says whether the values only ever go up (or only down) from one row to the next. Ages in a passenger list are "not monotonic"; a running number like `PassengerId` is increasing. For example, 1, 2, 5, 9 only goes up (increasing), while 3, 1, 2 goes down and then up (not monotonic).

The last two tabs deal with single values:

- **Common values:** the ages that occur most often. The top three are 24 (30 passengers), 22 (27) and 18 (26).
- **Extreme values:** the five smallest and five largest values. The smallest ages are 0.42, 0.67, 0.75, 0.75 and 0.83, all babies under a year; the largest are 80, 74, 71, 71 and 70.5.

The extreme values are where outliers show up first. Here they look genuine: babies and elderly passengers really were on board.

5. Interactions: a scatter plot for any pair

Key point: The Interactions section draws a scatter plot for any two numerical columns we pick, which is bivariate analysis on demand.

The Interactions section holds a scatter plot for every pair of numerical columns. Two rows of tabs pick the pair: the top row sets the x-axis column, the second row the y-axis column. Figure 8 shows `Age` on the x-axis against `Fare` on the y-axis.

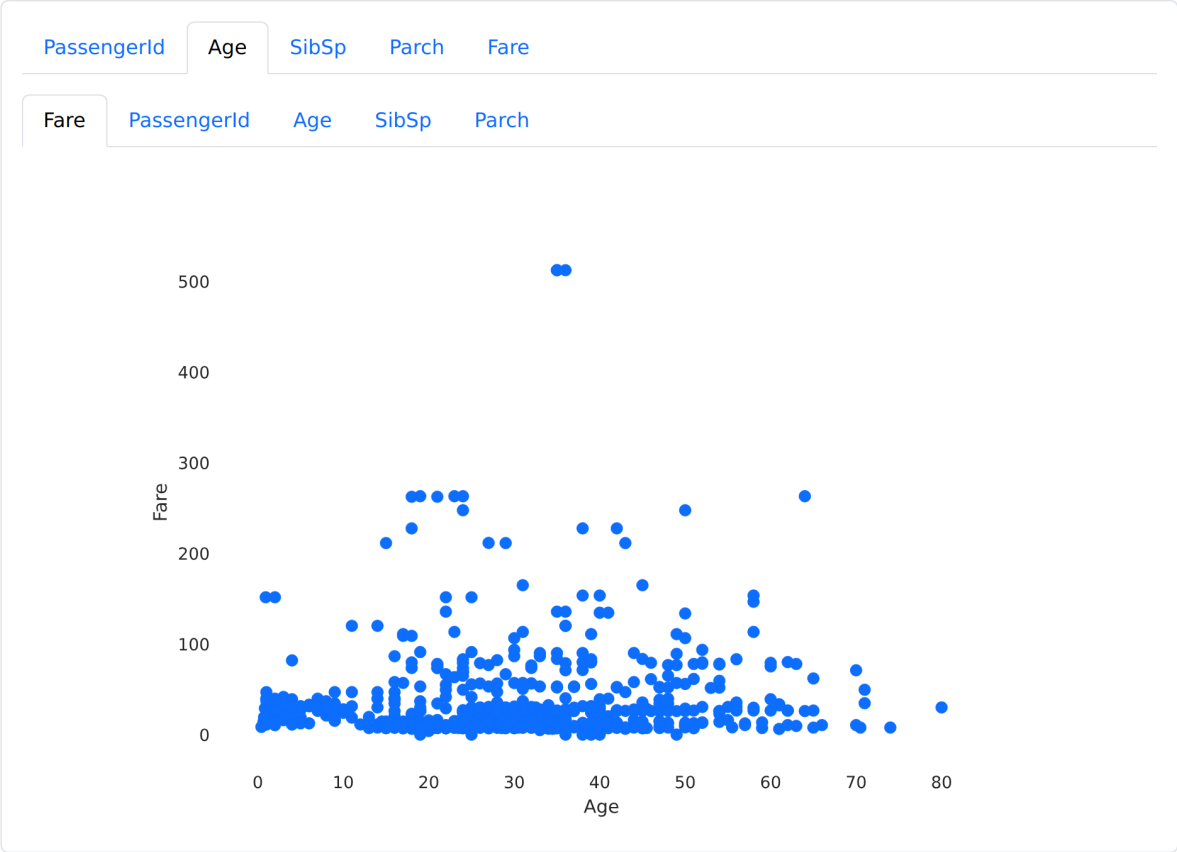


Figure 8: The Interactions section: Age against Fare

Most fares are below 100, at every age; the dots do not rise or fall with age, so the two columns are hardly related. A few fares stand far above the rest, the highest at 512. These are the extreme fares that the **box plot** (G-329) of `Fare` flagged as outliers in the fare box plot.

Picking the same column for both axes gives no useful plot: every dot would lie on one diagonal line. Only numerical columns appear here; categorical pairs are covered by the Correlations section.

6. Correlations: every pair in one heatmap

> **Key point:** The Correlations section shows the correlation of every pair of columns as a heatmap; dark blue is a strong positive link, dark red a strong negative one.

Correlation measures how strongly two columns move together, from -1 to $+1$, as defined in correlation. The Correlations section computes it for every pair and draws the results as a heatmap. Figure 9 shows **Pearson's r**, the coefficient for linear (straight-line) relationships

between numbers.

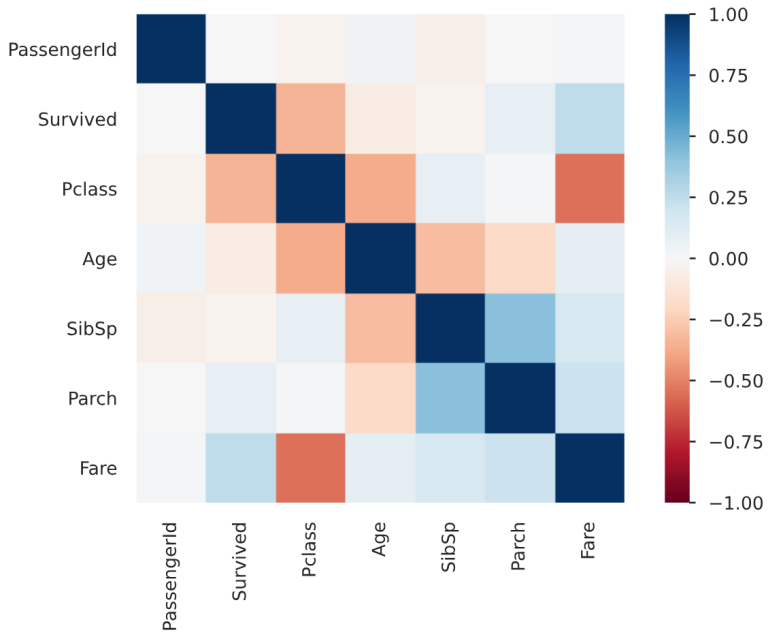


Figure 9: Pearson's r for every pair of numerical columns

Pearson's r needs numbers. `Survived` and `Pclass` are categorical, but their values are stored as numbers (0 and 1; 1, 2 and 3), so the heatmap includes them. `Sex` and `Embarked` are stored as text, so they are left out.

Each cell is the correlation of its row and its column. The colour bar on the right reads the colours: dark blue near +1, white near 0, dark red near −1. The diagonal is always dark blue, since each column matches itself perfectly.

Away from the diagonal, a few cells stand out:

- **`Pclass` and `Fare`:** −0.55**.** The strongest link. A higher class number (cheaper class) goes with a lower fare.
- **`SibSp` and `Parch`:** +0.41**.** Passengers with siblings or a spouse aboard often had parents or children aboard too: they travelled as families.
- **`Survived` and `Pclass`:** −0.34, and **`Survived` and `Fare`:** +0.26**.** Passengers in better classes, who paid more, survived more often.

So at a glance we learn which **features** (input variables, one column each) relate to the **target** (`Survived`, the output we predict), and which features relate to each other. Both matter later: features linked to the target are useful, and features strongly linked to each other repeat the same information.

Python: Choosing the coefficients.

```
profile = ProfileReport(df, correlations={
    "auto": {"calculate": True},
    "pearson": {"calculate": True},
})
```

By default only the "Auto" tab is computed. The Notebook switches on Pearson's r as well, as above. The same way, `"spearman"`, `"kendall"`, `"phi_k"` and `"cramers"` add more tabs.

Extra: Pearson's r only works for two numerical columns, and only sees linear (straight-line) links. Other coefficients cover the other cases:

- **Spearman's** and **Kendall's** coefficients measure whether two columns rise together at all, even along a curve, by comparing the ranks of the values instead of the values (Spearman 1904; Kendall 1938). Example: $x = 1, 2, 3, 4$ and $y = 1, 4, 9, 16$ lie on a curve. Their ranks are 1, 2, 3, 4 in both lists, identical, so Spearman's coefficient is exactly 1, while Pearson's r is 0.98 because the points are not on a straight line.
- **Cramér's V** measures the link between two categorical columns, from 0 (none) to 1 (complete) (Cramér 1946). For `Sex` and `Survived` the counts are 81 females who died, 233 females who survived, 468 males who died and 109 males who survived; the computer gives $V = 0.54$, a strong link.
- **Phik** (ϕ_k) works for any mix of numerical and categorical columns (Baak et al. 2020).

The default "Auto" heatmap picks a suitable coefficient for each pair: Spearman's for two numerical columns, and Cramér's V when a categorical column is involved. The Auto heatmap is how the report found the strong link between `Sex` and `Survived` in the alerts: `Sex` is text, so Pearson's r cannot include it at all. Every pair above 0.5 on the Auto heatmap gets a "High correlation" alert (0.5 is the library's default `threshold`).

Extra: Next to each Heatmap tab is a Table tab with the exact numbers behind the colours.

7. Missing values: how many, and where

Key point: The Missing values section shows, per column, how many values are present (Count) and in which rows the gaps are (Matrix).

The fifth section is all about missing values. Its first tab, Count (Figure 10), draws one bar per column; the bar's height is the share of rows that have a value.

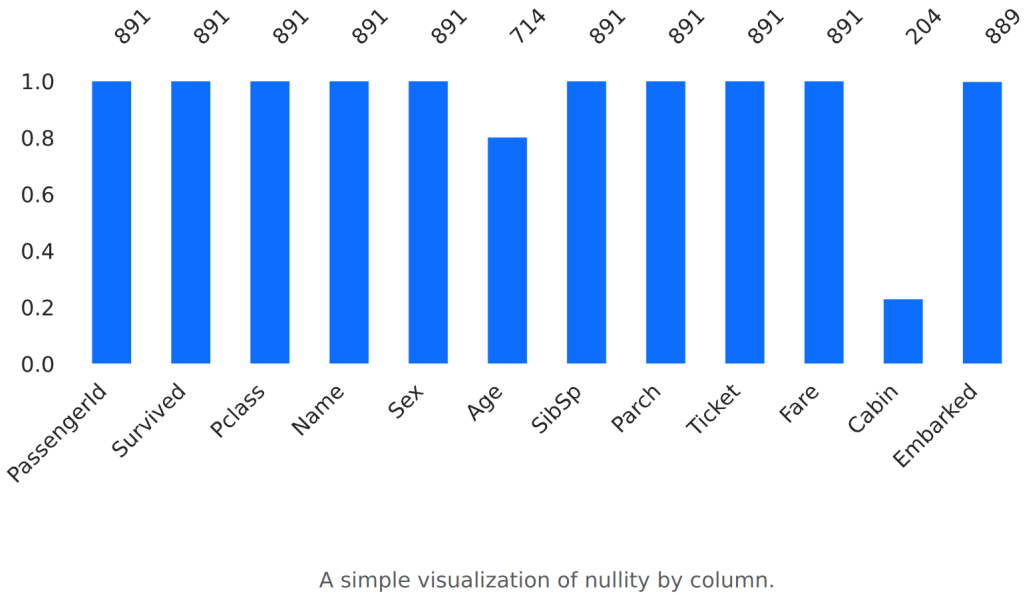


Figure 10: The Count tab of the Missing values section

The numbers above the bars are the counts of present values. Most columns are complete, with all 891. Three are not:

- **`Age:`** 714 present, so 177 missing.
- **`Cabin:`** only 204 present, so 687 missing.
- **`Embarked:`** 889 present, so 2 missing.

The second tab, Matrix (Figure 11), draws the whole table as a picture, a **nullity matrix** ("nullity" means missing). Each column of the data is a vertical strip, and each row of the data a thin horizontal line through all strips, from row 1 at the top to row 891 at the bottom. A filled line means the value is there; a white line means it is missing.

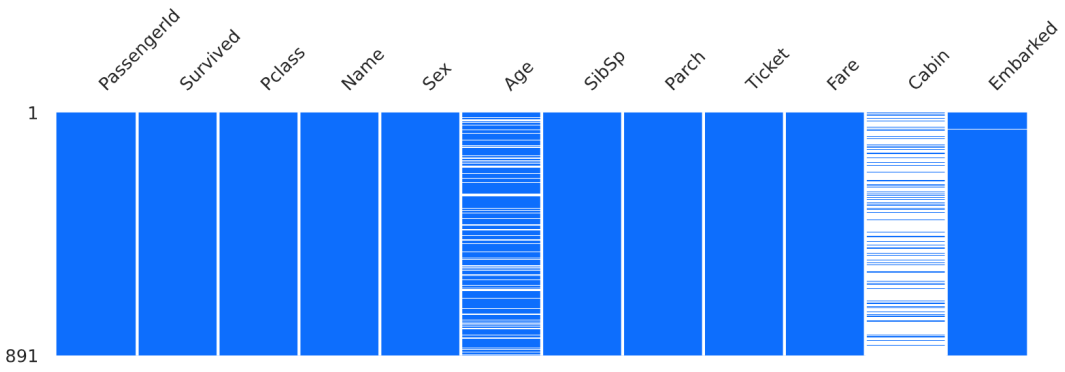


Figure 11: The Matrix tab of the Missing values section

The more white, the more missing values. `Cabin` is mostly white, `Age` has white lines scattered all the way down, and `Embarked` has just two: the one near the top (row 62) is visible, while the second, at row 830, is too thin to show at this size. The matrix also shows whether gaps in different columns fall in the same rows; here they do not line up in any clear pattern.

> **Extra:** The third tab, Heatmap, shows whether the gaps in one column tend to occur together with gaps in another. Missing values and how to handle them get their own Notes later, starting with why missing values must be handled.

8. Sample: a look at the rows

> **Key point:** The Sample section shows the first and last rows of the data, like `df.head()` and `df.tail()`.

The last section shows real rows, in two tabs: First rows and Last rows. They give a feel for what the values look like, as `df.head()` did in what the data looks like. The last rows also show whether the end of the file differs from the start, for example a total row or a different format.

9. Using the report

> **Key point:** We run a profiling report first on every new dataset, read it section by section, and write down what we notice.

The report covers in seconds what took the first questions, **univariate analysis** (G-2050) and bivariate and multivariate analysis by hand. The report does not replace understanding: it lists facts, and we decide what they mean. A good way to use it (Figure 12):

1. Build the report as soon as we get a new dataset.
2. Read it in order: Overview and Alerts, then each variable, then Interactions, Correlations and Missing values.
3. Write down every observation, for example "`Cabin` is 77% missing" or "`Pclass` and `Fare` are strongly linked".
4. Turn the observations into tasks: columns to drop, missing values to fill, outliers to check.

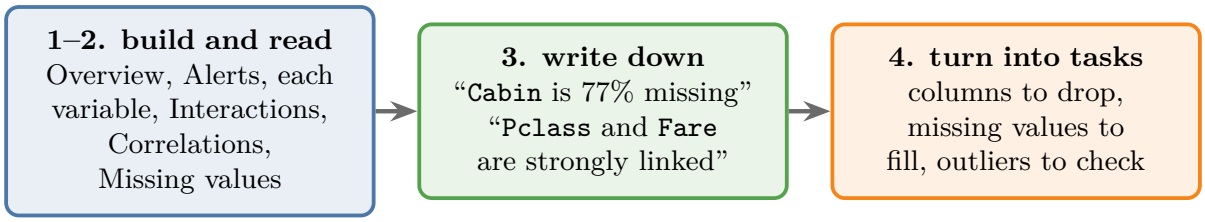


Figure 12: Using the report: read the sections in order, write down what we notice (the two examples above), and turn each observation into a task

Reading reports becomes faster with practice. Running the library on three or four different datasets, and writing down observations each time, builds the habit of knowing where to look.

> **Extra:** The report knows nothing about the meaning of the data. The report cannot tell that `Survived` is the target we want to predict, that a 0 fare might be a crew member or a free ticket, or that `PassengerId` is a label rather than a measurement. Those judgements still come from us, and from hand-made EDA when a question needs a closer look.

10. Summary

Section	What it shows	Hand-made equivalent	Titanic example	Why it matters
Overview	Size, types, missing cells, duplicates, memory	`shape`, `info`, `isnull`, `duplicated`	12 columns, 891 rows, 8.1% missing	Answers the first questions in one table
Alerts (in Overview)	Possible problems, one per line	Reading every column ourselves	`Cabin` 77.1% missing	Points in seconds to columns to clean or that look useless
Variables	Each column: statistics and a chart	Univariate analysis	Age: mean 29.70, IQR 17.875	Extreme values show where outliers appear first
Interactions	Scatter plot of any two numerical columns	Scatter plots	A few fares above 500	Bivariate analysis on demand, for any pair
Correlations	Correlation of every pair, as a heatmap	`df.corr()` and a heatmap	`Pclass` and `Fare`: -0.55	Shows which features relate to the target and which repeat each other
Missing values	Present values per column, and where the gaps are	`isnull().sum()`	`Age` 714, `Cabin` 204 of 891	Shows whether gaps in different columns fall in the same rows
Sample	First and last rows	`head`, `tail`		The last rows show whether the end of the file differs from the start

- `ProfileReport(df).to_file("report.html")` builds a full EDA report as one web page, so most of the hand-made EDA takes three lines and a few seconds.
- The library has had three names: `pandas-profiling`, `ydata-profiling`, and now `fg-data-profiling` (import `data_profiling`), so older tutorials still work after changing the name; the `ProfileReport` call is the same.
- Alerts point to columns to check: high correlation, missing, unique, uniform, zeros. Each alert is a question, not a verdict, because many zeros in `SibSp` is simply a fact

while 15 free tickets deserve a closer look.
- "Auto" correlations cover categorical columns too; Pearson's r only covers numbers. That is why the link between `Sex` and `Survived` shows up only on the Auto heatmap: `Sex` is text.
- `minimal=True` keeps the report fast on big data, because it skips the slow parts (correlations, interactions, missing-value charts).
- The report lists facts; we still read it, write down observations and decide what to do, because it knows nothing about the meaning of the data, such as which column is the target.
- So one call does most of the first questions, univariate and bivariate analysis by itself; turning its facts into tasks (columns to drop, gaps to fill, outliers to check) is still our job.

11. Sources

Built from

- CampusX, "Pandas Profiling | Day 22 | 100 Days of Machine Learning", YouTube, https://www.youtube.com/watch?v=E69Lg2ZgOxg

Other references

- Baak, M., Koopman, R., Snoek, H. and Klous, S. (2020). A new correlation coefficient between categorical, ordinal and interval variables with Pearson characteristics. *Computational Statistics & Data Analysis* 152.
- Cramér, H. (1946). *Mathematical Methods of Statistics*. Princeton University Press.
- Kendall, M. (1938). A New Measure of Rank Correlation. *Biometrika* 30(1/2).
- PyPI. pandas-profiling, ydata-profiling and fg-data-profiling project pages. pypi.org.
- Spearman, C. (1904). The Proof and Measurement of Association between Two Things. *American Journal of Psychology* 15(1).

12. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Pandas Profiling (G-1440)	A library that builds a full EDA report (column summaries, missing values, correlations and more) from a DataFrame in a few lines of code; now named `fg-data-profiling`.
Alert (G-186)	In a pandas profiling report, a warning about a column that may need attention, such as many missing values; a question to check, not a verdict.

Term	Meaning
Average record size (G-239)	The memory one row takes, on average.
High cardinality (G-896)	A categorical column with very many different categories.
Median absolute deviation (MAD) (G-1207)	The median distance of the values from their median.
Monotonicity (G-1260)	Whether a column's values only go up, or only go down, from row to row.
Nullity matrix (G-1364)	A picture of the whole table with each missing value drawn as a white line; it shows where values are missing and whether gaps in different columns fall in the same rows.
Profiling report (G-1580)	An automatic EDA report describing every column and pair of columns of a dataset.
Univariate analysis (G-2050)	Studying one column (variable) on its own, mostly with graphs, to see its values and spread; the column's type decides which graphs to draw.
HTML (G-904)	The language web pages are written in: a tree of nested tags.
PyPI (G-1593)	The Python Package Index, the public store of Python packages.
Count plot (G-494)	A bar chart with one bar per category, as tall as its frequency.
Box plot (G-329)	A graph of a column's five-number summary, a box with whiskers and outliers drawn as dots; it shows the centre, spread and outliers at a glance.
Skewness (G-1817)	A number for how lopsided a distribution is: 0 symmetric, positive right tail, negative left tail.
Variable (G-2071)	One column of a dataset.
Observation (G-1374)	One record of a dataset: one row of the data table.
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Target (G-1949)	The output column we predict, such as the class; also called the label.
Coefficient of variation (CV) (G-408)	Standard deviation divided by mean: spread relative to the average.
Kurtosis (G-1021)	How heavy the tails of a distribution are compared with a normal curve; the fourth moment.

Term	Meaning
Pearson correlation coefficient (Pearson's r) (G-1474)	A number from -1 to $+1$ that measures how closely two numerical columns follow a straight-line relationship ($+1$: they rise together exactly; 0: no straight-line link), written r; it is the one `df.corr()` computes.
Cramér's V (G-500)	A measure of how strongly two categorical features are related, from 0 (no relationship) to 1 (perfect relationship), made by rescaling the chi-square statistic.